Artificial Intelligence & Machine Learning Internship

Project Report

RISING WATER – AI-Based Flood Prediction System using Machine Learning

SmartBridge Educational Services Pvt. Ltd.

in collaboration with

AICTE (All India Council for Technical Education)

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ABSTRACT

Floods are among the most destructive natural disasters, causing severe damage to life, infrastructure, agriculture, and the economy. Accurate flood prediction enables governments and disaster management authorities to take preventive actions and reduce losses.

The **Rising Waters – AI-Based Flood Prediction System** is a web-based application that predicts flood risk using Machine Learning techniques. The system uses environmental parameters such as Monsoon Intensity, Drainage Systems, River Management, Deforestation, and Urbanization to determine whether a region is at high or low risk of flooding.

The dataset is preprocessed using StandardScaler, and an XGBoost Classifier is trained to perform binary classification. The trained model is integrated into a Flask web application, providing users with an interactive interface for entering environmental conditions and receiving instant flood risk predictions.

The application also displays prediction confidence, confidence level, safety recommendations, and maintains a prediction history for future reference.

This project demonstrates the practical application of Artificial Intelligence in disaster management and contributes toward building smarter and more resilient communities

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CHAPTER 1 INTRODUCTION

1.1 Introduction

Floods are one of the most common and devastating natural disasters occurring worldwide. Heavy rainfall, poor drainage systems, rapid urbanization, deforestation, and ineffective river management contribute significantly to flood occurrences. Every year, floods result in enormous economic losses, displacement of populations, and destruction of infrastructure.

Figure 1.1 Flood-affected area after heavy rainfall

Traditional flood forecasting methods primarily rely on historical observations and manual analysis, which may not provide timely or highly accurate predictions. Recent advancements in Artificial Intelligence and Machine Learning have enabled the development of intelligent prediction systems capable of analyzing complex environmental factors and providing reliable flood risk assessments.

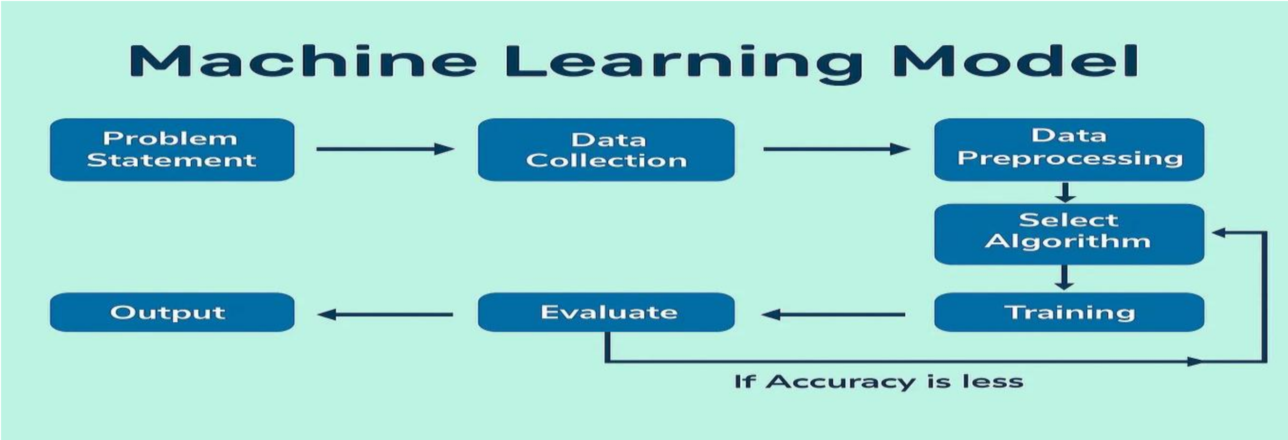


Figure 1.2 General Machine Learning Workflow

The Rising Waters project aims to develop an AI-powered flood prediction system capable of classifying flood risk based on selected environmental parameters. The system is designed to assist disaster management authorities, local governments, and citizens by providing quick predictions and safety recommendations.
The project integrates Machine Learning with web technologies, allowing users to interact with the prediction model through an easy-to-use web interface.

1.2 Problem Statement

Flood disasters often occur with little warning, causing extensive damage to life and property. Existing prediction systems are either expensive, complex, or require specialized infrastructure.
Therefore, there is a need for a simple, cost-effective, and intelligent flood prediction system that can analyze environmental conditions and provide timely predictions to support disaster preparedness and mitigation.

1.3 Objectives

The main objectives of the project are:

- Develop an AI-based flood prediction model.
- Predict flood risk using environmental parameters.
- Provide confidence scores for predictions.

- Display safety recommendations based on prediction results.
- Maintain prediction history.
- Build a responsive web application using Flask.

1.4 Scope of the Project

The project focuses on predicting flood risk using five major environmental parameters.

The developed system can be used by:

- Disaster Management Authorities
- Local Governments
- Environmental Researchers
- Educational Institutions
- General Public

The system can also serve as a foundation for future integration with IoT sensors, weather APIs, and GIS-based flood monitoring systems.

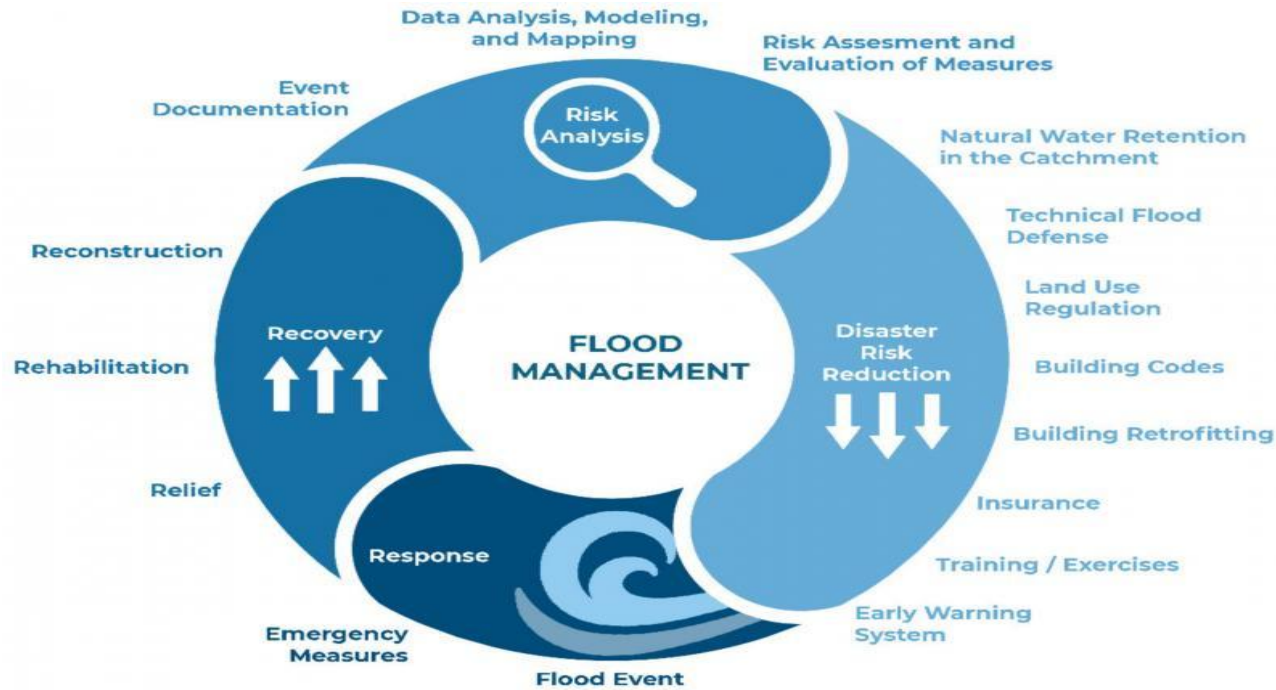


Figure 1.3 Flood Disaster Management Process

CHAPTER 2 LITERATURE SURVEY

2.1 Introduction

Flood prediction has become an important research area due to the increasing frequency of natural disasters worldwide. Researchers have proposed various techniques ranging from traditional statistical methods to advanced Artificial Intelligence and Machine Learning algorithms. Conventional flood prediction systems mainly rely on rainfall observations, river water levels, and historical flood records. Although these methods provide useful insights, they often fail to capture complex relationships among multiple environmental factors.

Machine Learning techniques overcome these limitations by analyzing large datasets and identifying hidden patterns that influence flood occurrence. Algorithms such as Decision Trees, Random Forests, Support Vector Machines, Artificial Neural Networks, and XGBoost have been widely adopted because of their high prediction accuracy and ability to handle nonlinear relationships.

The proposed Rising Waters system adopts the XGBoost algorithm due to its superior performance, computational efficiency, and robustness. The trained model is integrated with a Flask-based web application, enabling users to obtain real-time flood risk predictions using five important environmental parameters.

Table 2.1 Literature Survey

Method Used	**Advantages**	**Limitations**
Decision Tree	Simple Implementation	Lower Prediction Accuracy
Random Forest	Improved Accuracy	Higher Computational Cost
Artificial Neural Network	Learns Complex Relationships	Requires Large Datasets

XGBoost	High Accuracy and Prediction	Requires Parameter Tuning
XGBoost + Flask	High Accuracy,web-based interface	Uses only five selected features

Algorithm	Training Speed	Prediction Speed	Memory Usage	Scalability
Linear Regression	★★★★★ Very Fast	★★★★★ Very Fast	★★★★★ Low	★★★★ Excellent
Logistic Regression	★★★★★ Very Fast	★★★★★ Very Fast	★★★★★ Low	★★★★ Excellent
Decision Tree	★★★★ Fast	★★★★★ Very Fast	★★★★ Low	★★★ Good
Random Forest	★★★ Moderate	★★★ Moderate	★★ High	★★★★ Good
Naive Bayes	★★★★★ Very Fast	★★★★★ Very Fast	★★★★★ Low	★★★★ Excellent
KNN	★★★★★ Very Fast	★★ Slow	★★★★ Low	★★ Poor
SVM	★★ Slow	★★★ Moderate	★★★ Moderate	★★ Poor

Figure 2.1 Comparison of Machine Learning Algorithms Used for Flood Prediction

2.2 Existing System

Existing flood prediction systems mainly depend on historical rainfall records, river water levels, and manual analysis. Most traditional systems require extensive infrastructure and expert interpretation, making them difficult to deploy in remote or resource-limited regions. Furthermore, many systems do not provide prediction confidence or user-friendly interfaces.

2.3 Proposed System

The proposed Rising Waters system utilizes an XGBoost Classifier trained on flood-related environmental data. The application accepts five environmental parameters as input, preprocesses them using StandardScaler, and predicts the flood risk. The prediction result is displayed along with the confidence percentage, confidence level, risk level, and safety recommendations through a Flask web application.

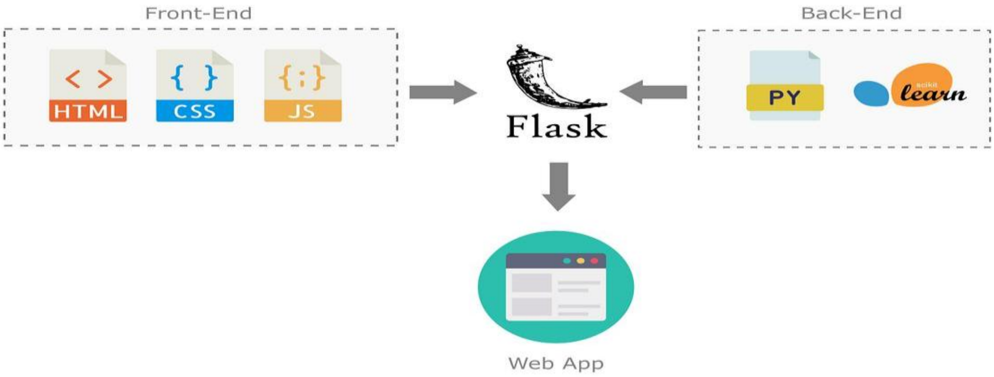


Figure 2.2 Proposed AI-Based Flood Prediction System

CHAPTER 3 IDEATION PHASE

3.1 Problem Identification

Flood disasters significantly affect human lives, agriculture, transportation, and infrastructure. In many cases, warning systems fail to provide timely and accurate predictions because they rely heavily on manual monitoring or limited environmental parameters.

An intelligent flood prediction system capable of analyzing multiple environmental factors can significantly improve disaster preparedness and reduce economic losses.

3.2 Proposed Solution

The proposed solution is the **Rising Waters** web application, which predicts flood risk using Machine Learning. The application uses five environmental parameters:

- Monsoon Intensity
- Drainage Systems
- River Management
- Deforestation
- Urbanization

The model processes these inputs and generates an immediate prediction along with a confidence score.

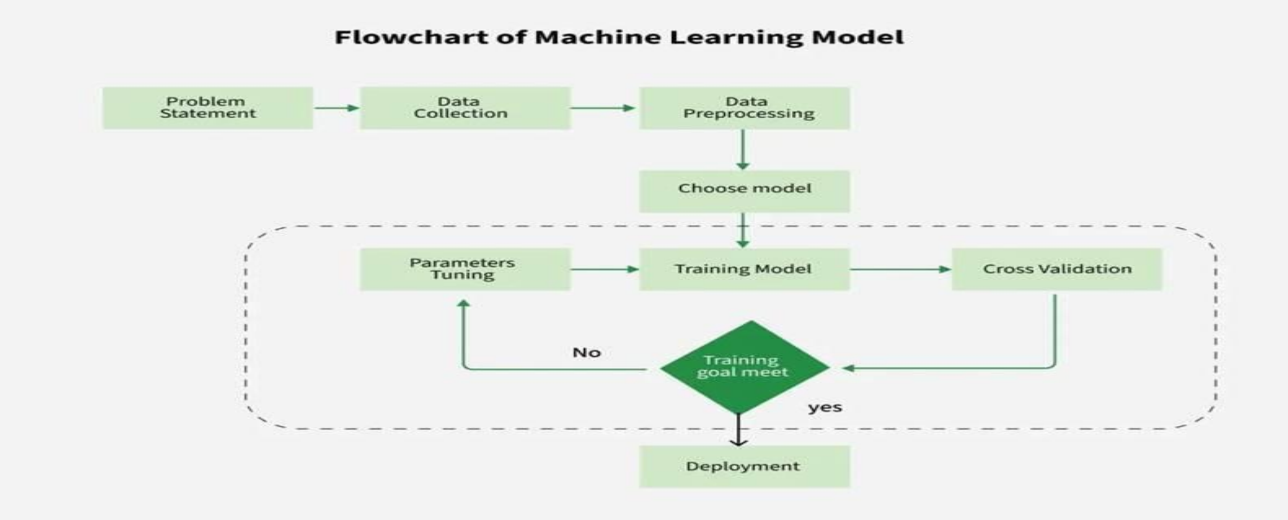


Figure 3.1 Project Ideation Flow

3.3 Objectives

The major objectives of the project include:

- Develop a flood prediction model using Machine Learning.
- Build a responsive Flask web application.
- Display prediction confidence.
- Maintain prediction history.
- Provide safety recommendations.
- Improve disaster preparedness.

3.4 Benefits

The developed system offers several benefits:

- Faster prediction.
- Easy-to-use interface.
- Reduced manual effort.
- Improved disaster management.
- Better public awareness.
- Cost-effective implementation.

CHAPTER 4 REQUIREMENT ANALYSIS

4.1 Functional Requirements

The application should provide the following functionalities:

- User input interface.
- Data preprocessing.
- Flood prediction.
- Confidence calculation.
- Prediction history.
- Safety recommendations.
- Navigation between pages.

4.2 Non-Functional Requirements

The system should satisfy the following requirements:

- High accuracy.
- Fast response time.
- User-friendly interface.
- Secure implementation.
- Reliable predictions.
- Easy maintenance.
- Cross-platform compatibility.

4.3 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or above
RAM	4 GB Minimum
Storage	20 GB Free Space
Internet	Required for GitHub deployment

4.4 Software Requirements

Software	**Version**
Operating System	Windows 10/11
Python	3.11 or above
Flask	Latest
XGBoost	Latest
VS Code	Latest
Git	Latest
Bootstrap	Version 5

4.5 Development Tools

Tool	**Purpose**
Python	Programming
Flask	Backend Framework
HTML	Web Pages
CSS	Styling
Bootstrap	Responsive Design
Pandas	Data Processing
NumPy	Numerical Computation
Scikit-learn	Data Scaling
XGBoost	Machine Learning Model
Joblib	Model Serialization
GitHub	Version Control

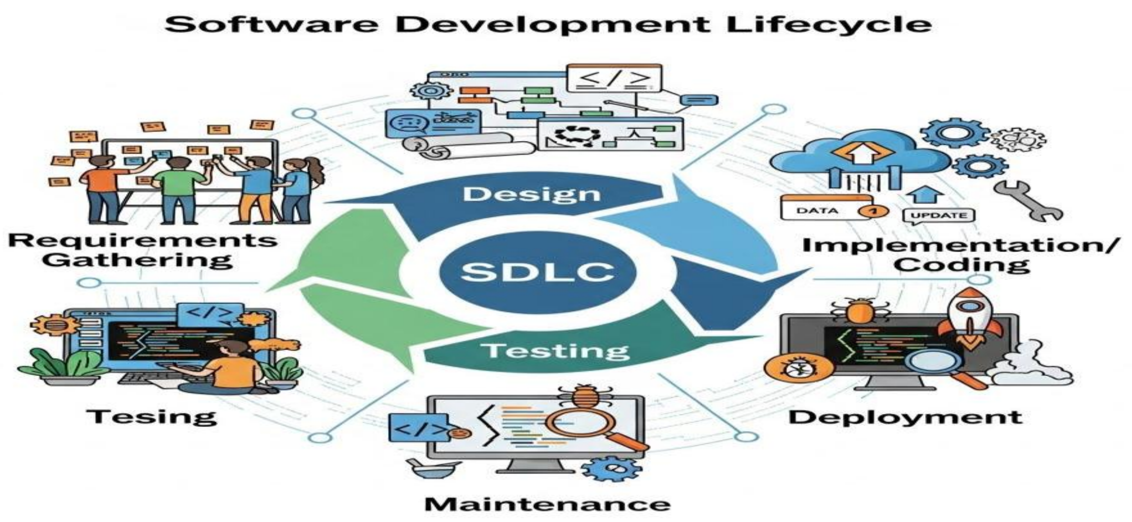


Figure 4.1 Software Development Process

CHAPTER 5 PROJECT DESIGN

5.1 System Architecture

The Rising Waters – AI-Based Flood Prediction System follows a modular architecture that integrates data preprocessing, machine learning prediction, and a web-based user interface. The system is designed using the Flask framework and the XGBoost classification algorithm.

The architecture consists of the following components:

1. User Interface
2. Flask Backend
3. Data Preprocessing Module
4. Machine Learning Prediction Model
5. Result Generation Module
6. Prediction History Module

The user enters five environmental parameters through the web interface. The backend validates the inputs and passes them to the StandardScaler for preprocessing. The scaled data is then sent to the trained XGBoost model, which predicts the flood risk. Finally, the application displays the prediction, confidence score, confidence level, and safety recommendations.

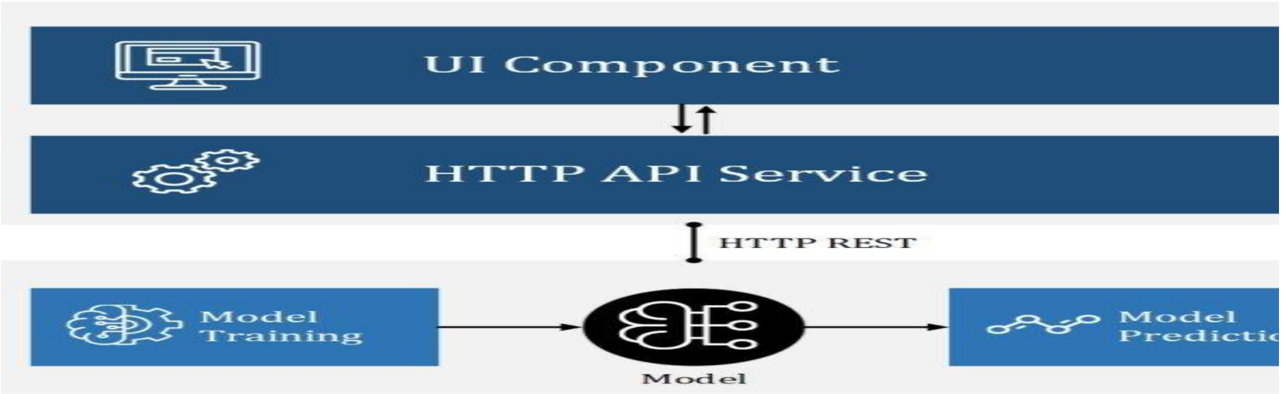


Figure 5.1.1 System Architecture

User
↓
HTML Form (Flask Frontend)
↓
Flask Backend
↓
Input Validation
↓
StandardScaler
↓
XGBoost Classifier
↓
Prediction + Confidence Score
↓
Result & Recommendations
↓
Prediction History (CSV)

Figure 5.1.2 Overall System Architecture

5.2 System Modules

The project consists of six major modules.

Module 1 – User Interface

This module provides a user-friendly web interface where users enter environmental parameters required for flood prediction.
Features:

- Responsive design
- Bootstrap interface
- Input validation
- Navigation pages

Module 2 – Data Preprocessing
The preprocessing module converts user input into numerical format and scales the values using the StandardScaler before prediction.
Functions:

- Data validation
- Standardization
- Feature preparation

Module 3 – Prediction Module
This module loads the trained XGBoost model and predicts flood risk.
Outputs:

- High Flood Risk
- Low Flood Risk

Module 4 – Confidence Estimation
The confidence module calculates prediction confidence using:
Confidence = max(predict_proba()) × 100
Confidence Levels

Confidence	Level
90–100	Very High
75–89	High
60–74	Moderate
Below 60	Low

Module 5 – Recommendation Module

Depending on prediction, the system displays safety recommendations.

High Risk:

- Prepare evacuation plans
- Issue flood alerts
- Monitor rainfall
- Deploy emergency teams

Low Risk:

- Continue monitoring
- Maintain drainage systems
- Follow weather advisories

Module 6 – History Module

Stores every prediction in CSV format.

Stored Information:

- Date
- Prediction
- Confidence

5.3 Use Case Diagram

Actors

- User

Use Cases

- Enter Parameters
- Predict Flood Risk
- View Result
- View History
- Clear History

Figure 5.2 Use Case Diagram

User

|

↓ ↓ ↓

Enter Data **Predict Risk** **View History**

↓

View Result

↓

Safety Recommendation

5.4 Activity Diagram

Start

↓

Open Application

↓

Enter Inputs

↓

Validate Input

↓

Scale Data

↓

Predict

↓

Display Result

↓

Store History

↓

End

5.5 Data Flow Diagram (DFD)

User

↓

Input Parameters

↓

Flask Application

↓

StandardScaler

↓

XGBoost Model

↓

Prediction

↓

Result Page

↓

History File

CHAPTER 6 PROJECT PLANNING & SCHEDULING

6.1 Planning

The project was planned in several stages to ensure systematic development.

The development activities included:

- Requirement Gathering
- Dataset Collection
- Data Preprocessing
- Feature Selection
- Model Training
- Model Evaluation
- Web Development
- Testing

- Documentation

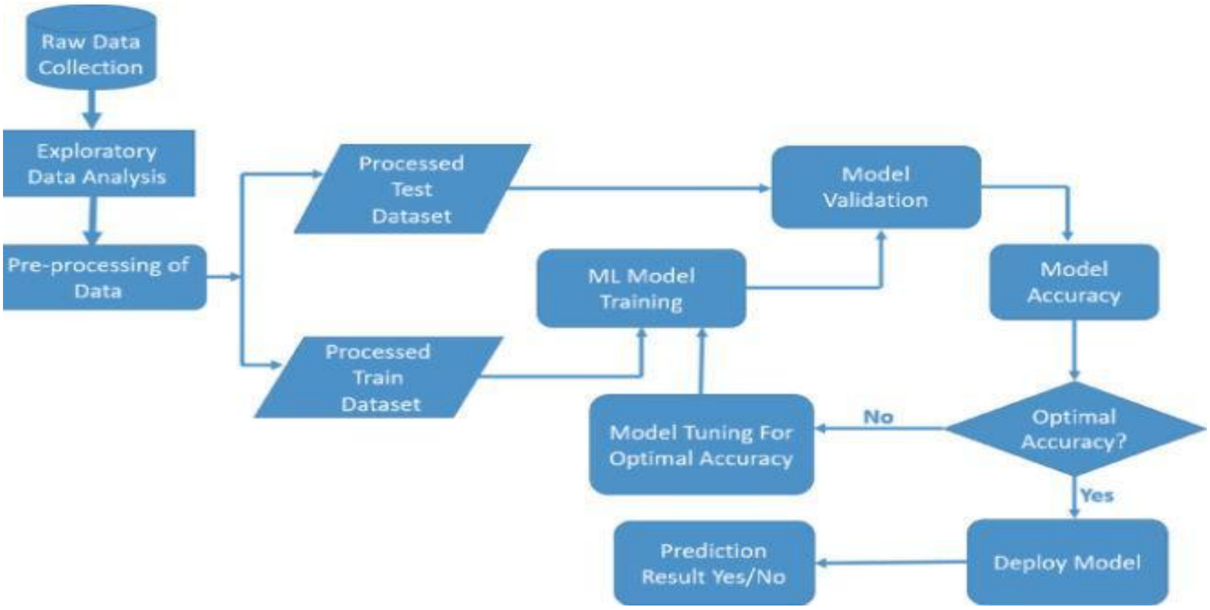
CHAPTER 7 METHODOLOGY

7.1 Overview

The proposed methodology follows a Machine Learning pipeline from data collection to prediction.

The major steps are:

- Data Collection
- Data Cleaning
- Feature Selection
- Data Scaling
- Model Training
- Model Testing
- Deployment
- Prediction



7.2 Methodology Flow

Dataset

↓

Preprocessing

↓

Feature Selection

↓

Train-Test Split

↓

StandardScaler

↓

XGBoost Training

↓

Model Evaluation

↓

Save Model

↓

Flask Deployment

↓

Prediction

7.3 Dataset Description

The flood dataset contains environmental factors affecting flood occurrence.

Selected Features:

Feature	Description
MonsoonIntensity	Rainfall intensity
DrainageSystems	Drainage condition
RiverManagement	River maintenance quality
Deforestation	Forest loss level
Urbanization	Urban development level

Target:

Flood Probability

7.4 Data Preprocessing

The preprocessing stage includes:

- Missing value checking
- Feature extraction
- Standardization
- Dataset splitting

Training Ratio
80%
Testing Ratio
20%

7.5 Machine Learning Algorithm

XGBoost Classifier

The XGBoost algorithm is selected because:

- High Accuracy
- Fast Prediction
- Handles Nonlinear Data
- Reduces Overfitting
- Efficient Memory Usage

7.6 Evaluation Metrics

The model performance is evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

CHAPTER 8 IMPLEMENTATION

8.1 Overview

The implementation phase converts the proposed system into a functional web application. The project is developed using Python and Flask, with the XGBoost machine learning model integrated into the backend. The application accepts user input through a web interface, preprocesses the data using StandardScaler, predicts the flood risk, and displays the result with confidence and safety recommendations.

The application consists of the following modules:

- Home Page
- About Page
- Prediction Module
- Result Page
- History Module

8.2 Home Page

The Home Page acts as the entry point of the application. It provides input fields for users to enter the five environmental parameters required for prediction.

Features

- User-friendly interface
- Responsive design
- Input validation
- Navigation bar
- Predict

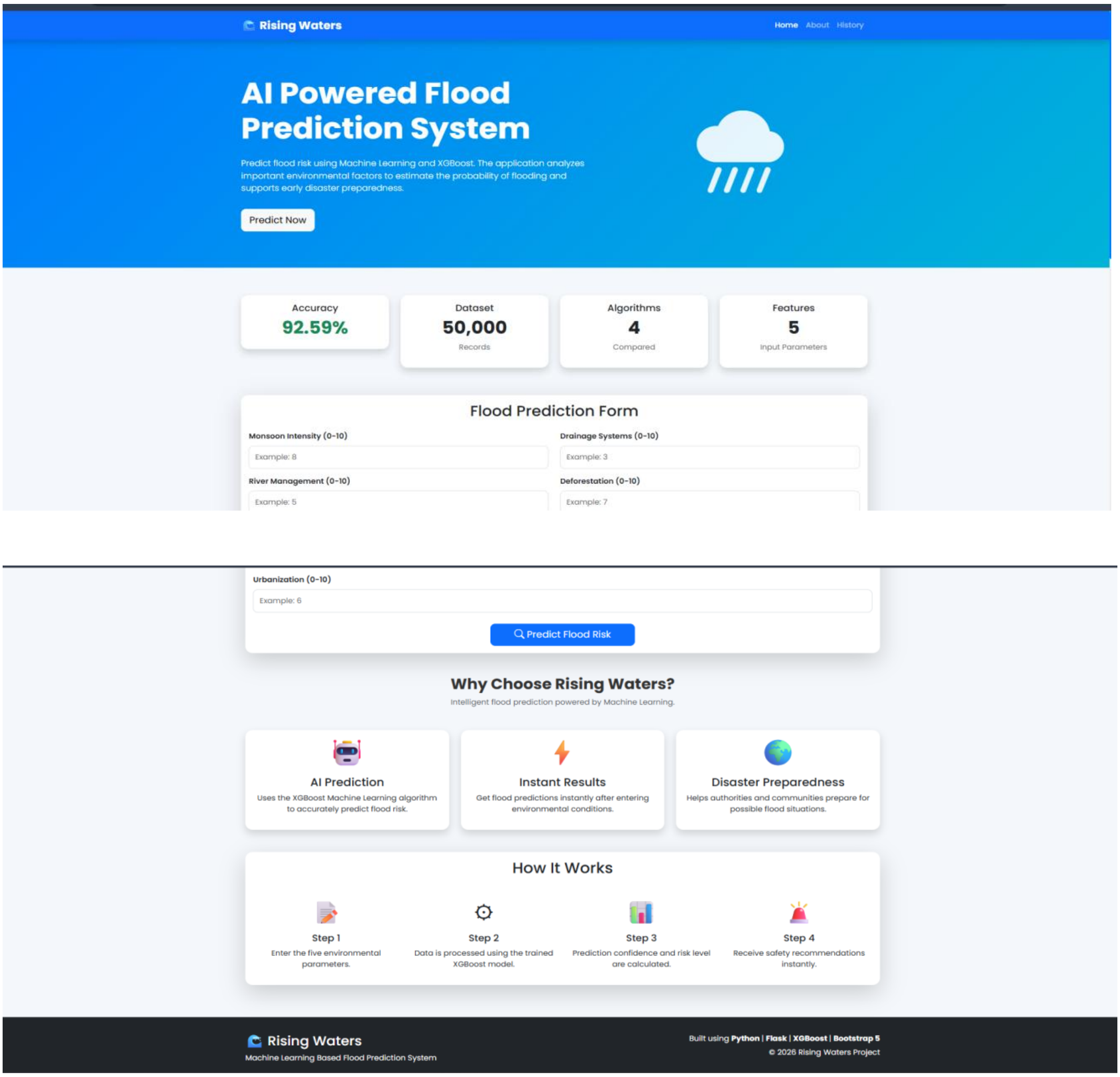


Figure 8.1 Home Page

8.3 About Page

The About page provides information about the project, objectives, technologies used, and system workflow.

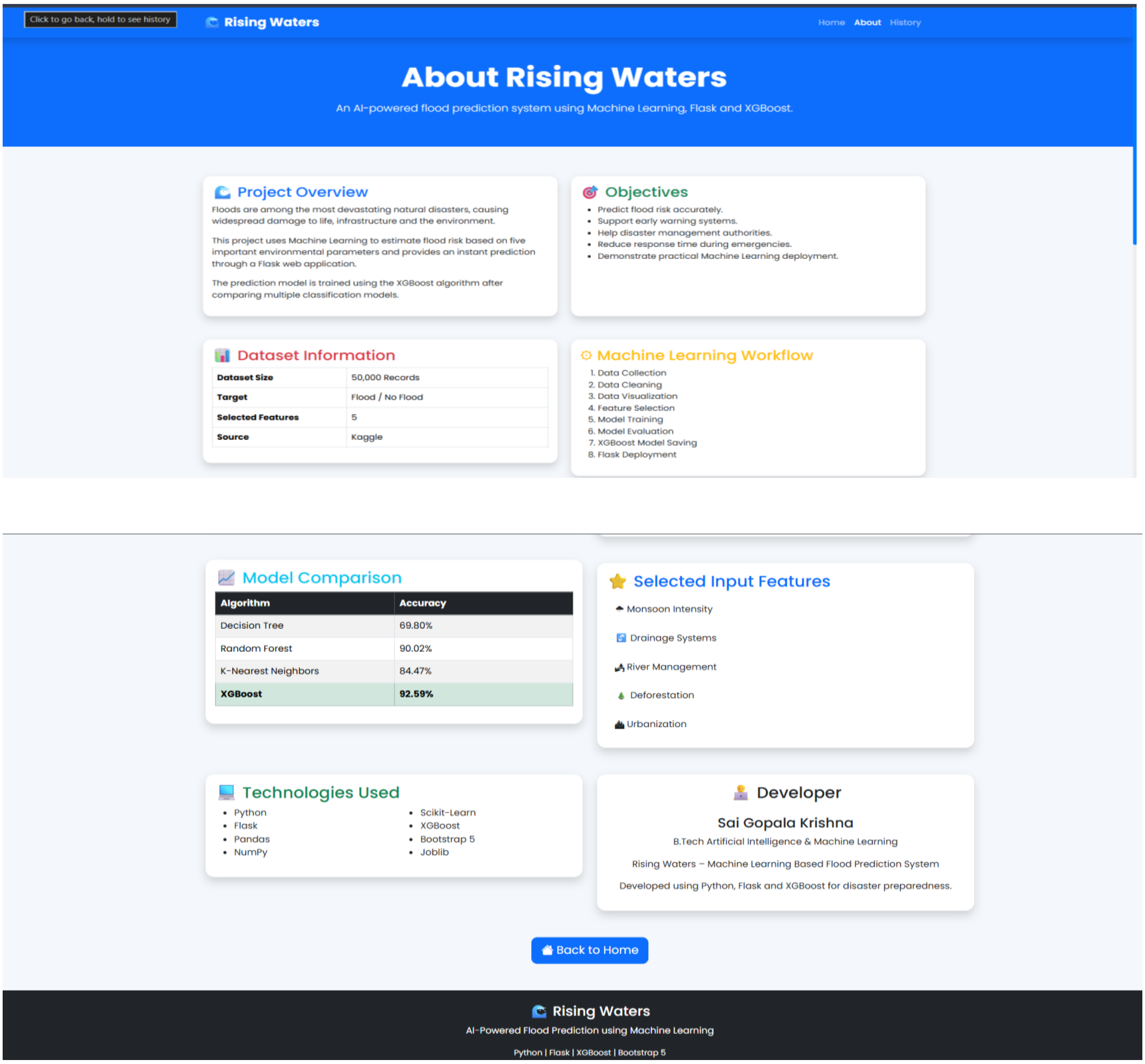


Figure 8.2 About Page

8.4 Prediction Module

The prediction module performs the following operations:

1. Reads user inputs
2. Validates input values
3. Scales the input using StandardScaler
4. Predicts flood risk using the trained XGBoost model
5. Calculates prediction confidence
6. Displays safety recommendations

8.5 Result Page

The Result page displays:

- Flood Risk
- Confidence Percentage
- Confidence Level
- Risk Level
- Safety Recommendations

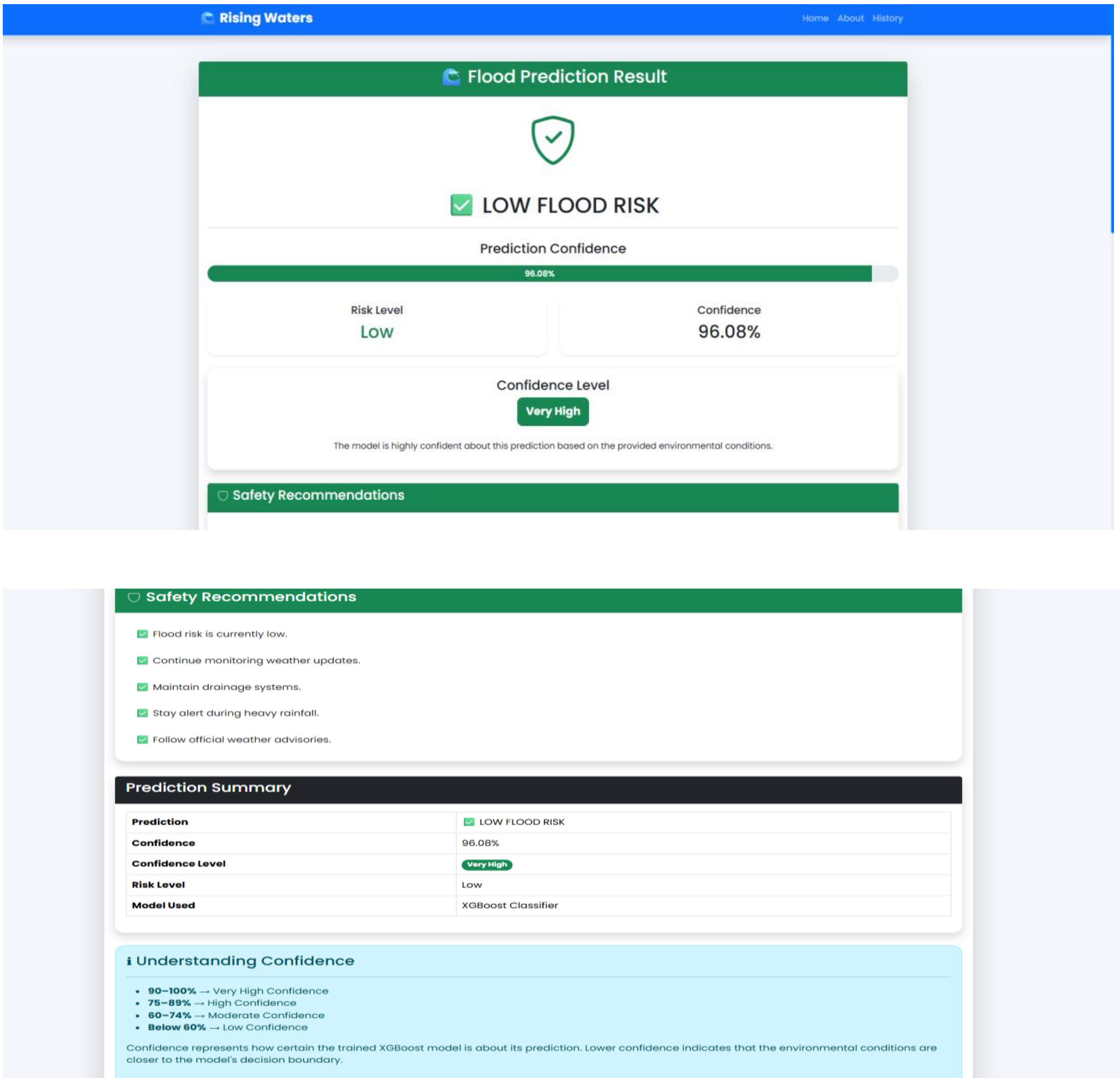


Figure 8.3 Prediction Result

8.6 History Module

Every prediction is stored in a CSV file for future reference.

Stored details include:

- Date
- Prediction
- Confidence

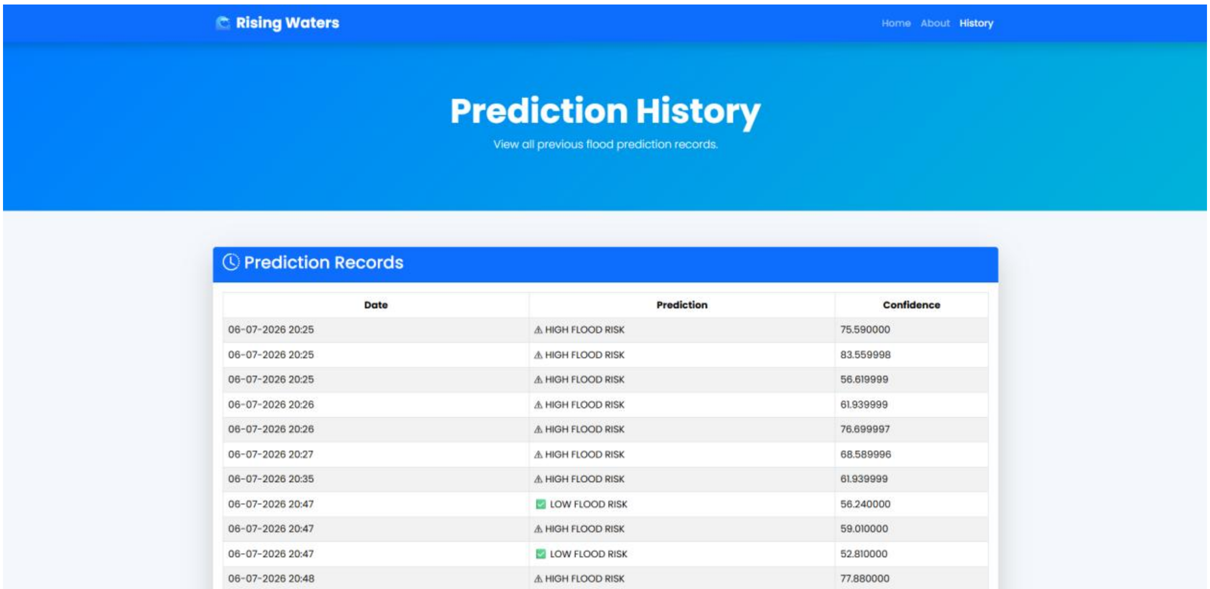


Figure 8.4 Prediction History

CHAPTER 9 TESTING

9.1 Introduction

Testing is carried out to ensure that the developed system functions correctly under different input conditions. The application is tested using various combinations of environmental parameters.

9.2 Test Cases

Test Case	Input	Expected Result	Status
TC-01	Low rainfall, good drainage	Low Flood Risk	Pass
TC-02	High rainfall, poor drainage	High Flood Risk	Pass
TC-03	Moderate values	Prediction Generated	Pass
TC-04	Invalid input	Validation Message	Pass
TC-05	View History	History Displayed	Pass

9.3 Performance Testing

Parameter	Result
Prediction Time	Less than 1 second
Accuracy	Based on trained XGBoost model
Response Time	Fast
Interface	Responsive
Stability	Stable

9.4 Validation

The system successfully:

- Predicts flood risk
- Calculates confidence
- Displays recommendations
- Stores prediction history
- Provides responsive user interaction

CHAPTER 10 RESULTS

10.1 Output Analysis

The developed system successfully predicts flood risk based on five environmental parameters.

The prediction output consists of:

- Flood Risk
- Confidence Score
- Confidence Level
- Safety Recommendations

Sample Inputs

Monsoon	Drainage	River	Deforestation	Urbanization	Prediction
2	9	9	2	2	Low Risk
9	2	2	9	9	High Risk
5	5	5	5	5	Depends on Model

10.2 Model Evaluation

The XGBoost model was selected due to its superior performance among the evaluated algorithms.

Performance Metrics

Metric	Value
Algorithm	XGBoost
Prediction Type	Binary Classification
Confidence	Probability-based
Deployment	Flask Web Application

CHAPTER 11 ADVANTAGES AND LIMITATIONS

Advantages

- Fast prediction
- High performance using XGBoost
- User-friendly interface
- Responsive web application
- Prediction confidence displayed
- Prediction history maintained
- Easy deployment
- Cost-effective solution

Limitations

- Uses only five environmental parameters
- Depends on the quality of the training dataset
- No real-time weather API integration
- No GIS map visualization
- Limited to binary classification

CHAPTER 12 CONCLUSION

Flood prediction plays an important role in minimizing the impact of natural disasters. The Rising Waters project demonstrates how Machine Learning can be effectively applied to disaster management using environmental data.

The developed system successfully predicts flood risk using five selected environmental parameters and provides users with confidence scores, safety recommendations, and prediction history. The integration of Flask with XGBoost enables efficient deployment through a user-friendly web interface.

Overall, the project meets its objectives and serves as a scalable foundation for future intelligent flood monitoring systems.

CHAPTER 13 FUTURE SCOPE

Several enhancements can be incorporated into the system in future versions:

- Integration with real-time weather APIs
- GIS-based flood visualization
- IoT sensor integration
- SMS and Email alerts
- Android and iOS mobile application
- Cloud deployment
- Satellite imagery analysis
- Deep Learning-based flood forecasting
- Multi-class flood severity prediction

REFERENCES

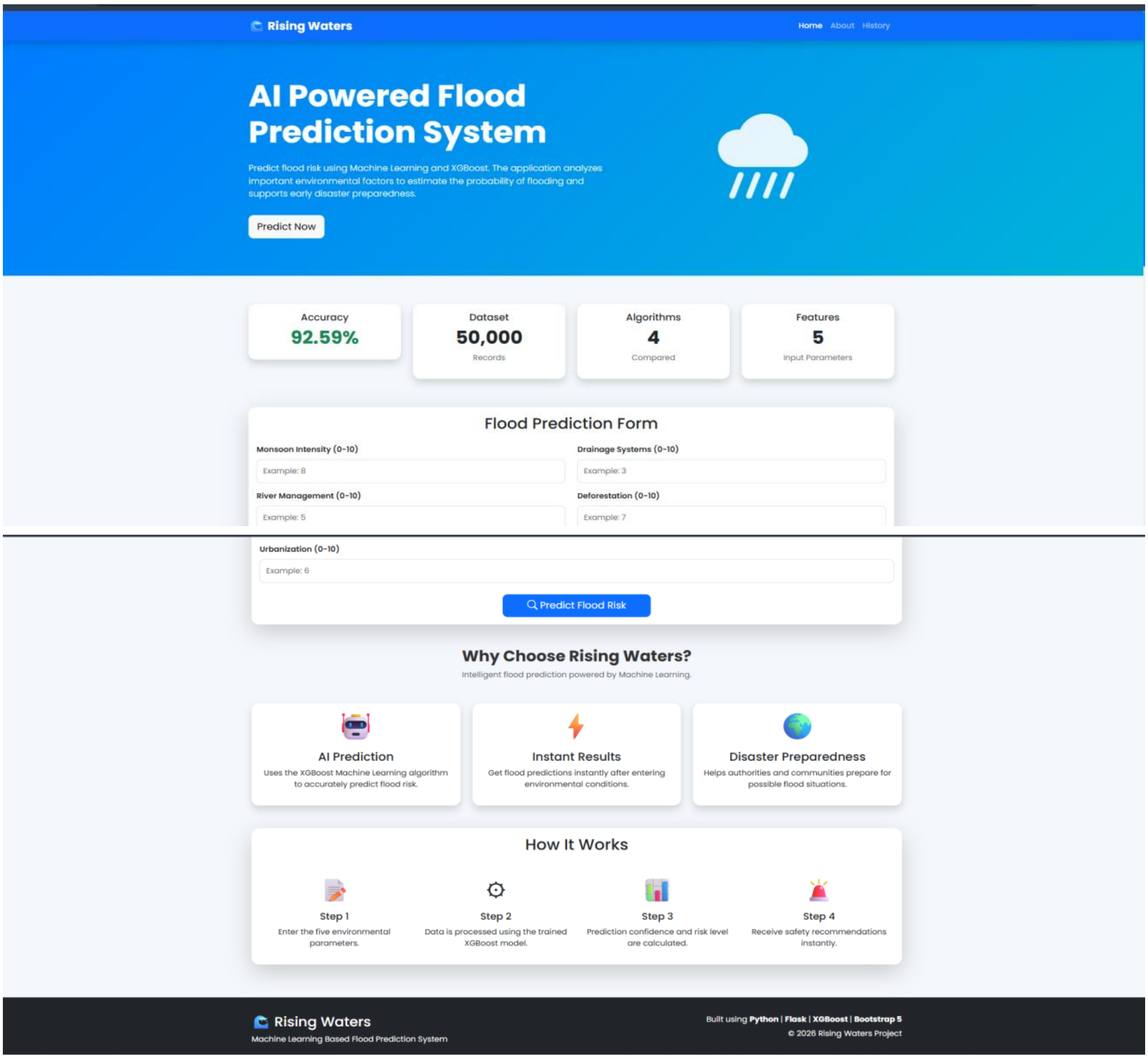
1. Scikit-learn Documentation.
2. Flask Official Documentation.
3. Python Official Documentation.
4. XGBoost Documentation.
5. Bootstrap 5 Documentation.
6. Pandas Documentation.
7. NumPy Documentation.

APPENDIX

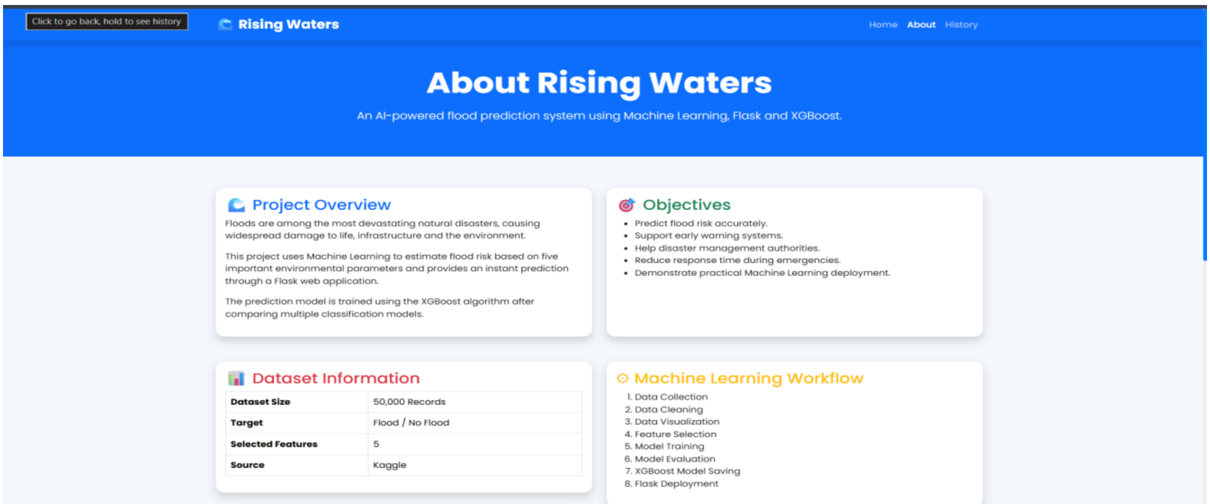
Project Structure

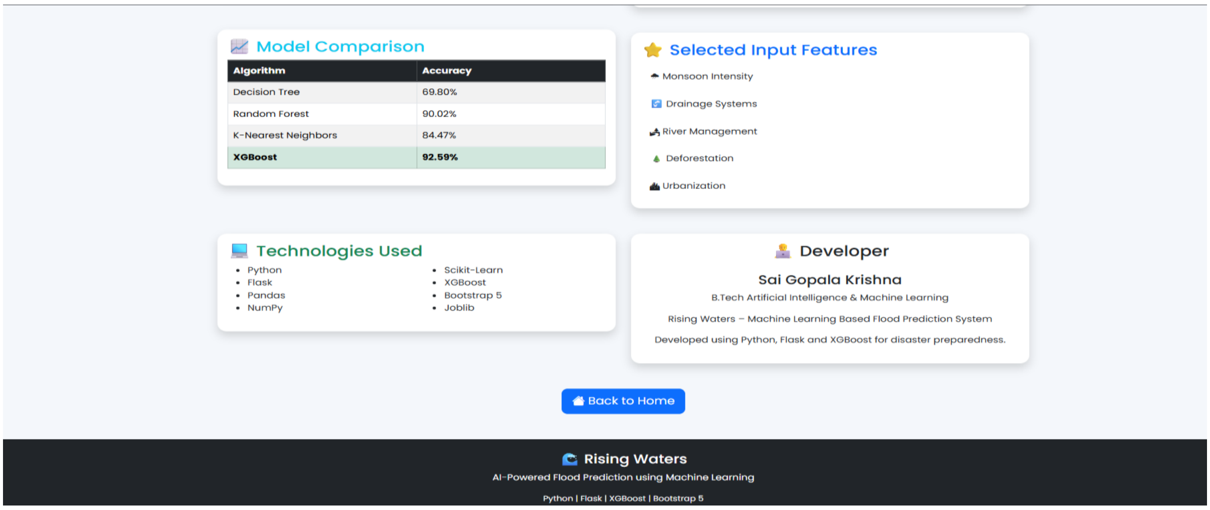
```
Rising-Waters/
│
├── app.py
├── datasets/
│   └── flood.csv
├── models/
│   ├── floods.save
│   └── scaler.save
├── templates/
│   ├── index.html
│   ├── about.html
│   ├── result.html
│   └── history.html
├── static/
│   └── style.css
├── data/
│   └── prediction_history.csv
├── notebook/
│   └── FloodPrediction.ipynb
├── Project Report.pdf
├── Demo.mp4
└── README.md
```

Screen Shots

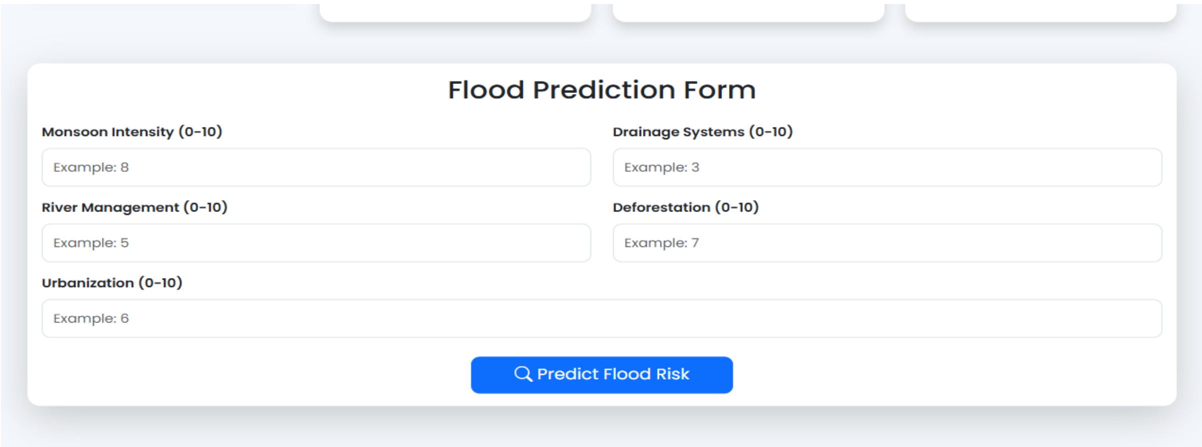


HOME PAGE

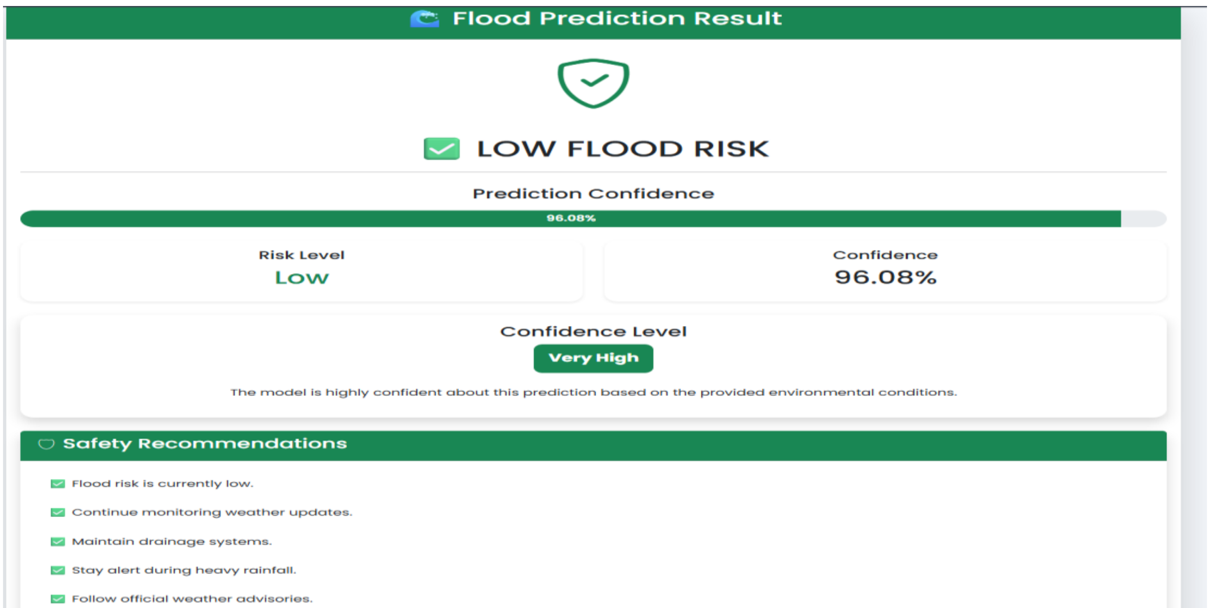




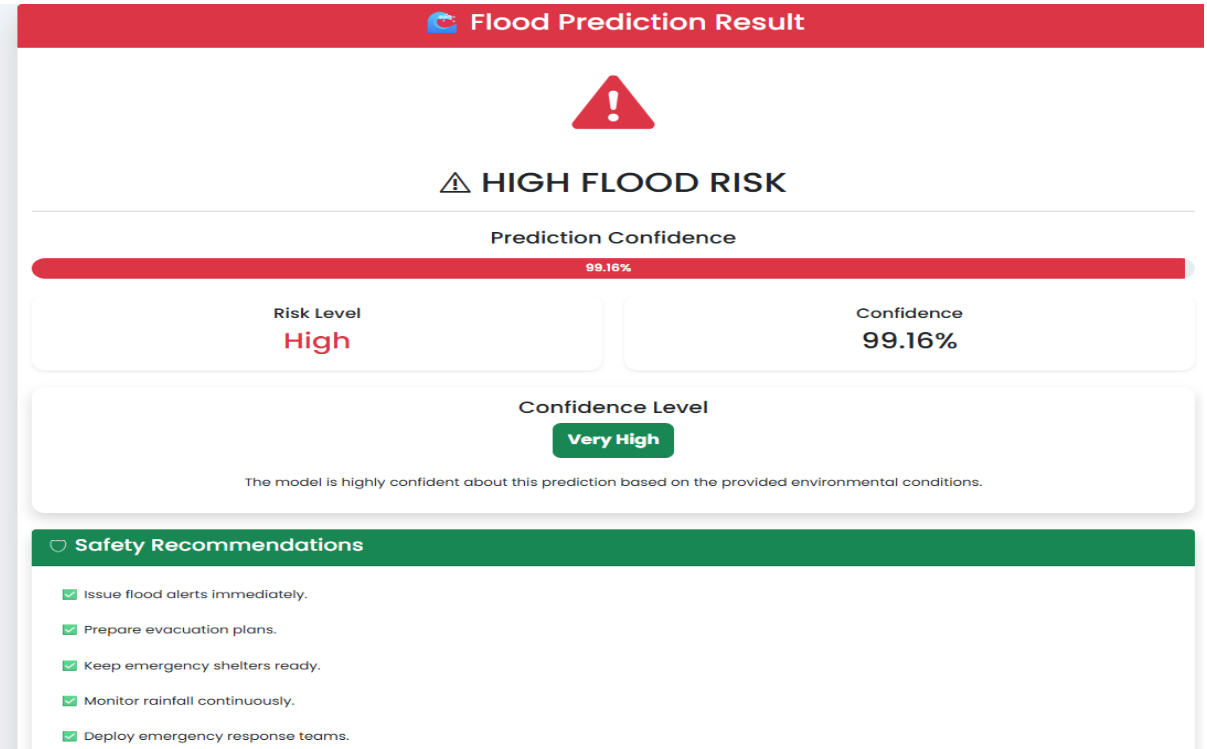
ABOUT PAGE



INPUT FORM



LOW FLOOD RISK RESULT



HIGH FLOOD RISK RESULT

GitHub Repository

Repository Name:
Rising-Waters

Repository URL:
https://github.com/Ajay12356789/Rising-Waters